

**ECON 101—Introduction to Microeconomics
Fall 2023**

Final Exam

December 4, 2023

Name:

NU ID number:

Instructions:

- This is a multiple choice exam, with 120 total points. Each question is worth X points. You will be provided scratch paper for your preliminary calculations but your official answers should be written **in the answer boxes below**. Answers written in scratch paper or outside the space provided will not be considered!
- The exam has a duration of 2 hours, from 12:00 to 14:00.
- You can request a bathroom break, but only one person can be out at any given time.
- On your desk, you are allowed to have your ID, writing instruments, a ruler, and a simple calculator (no graphing calculators). You may also have a simple watch if needed (smartwatches are not allowed).
- Please place all other personal items, including smart devices, to the side of the room.
- If you complete the exam early, you may leave the room, but not before 13:45. Otherwise, please remain in the room quietly until the exam concludes.

Multiple choice. Please write your answers here!

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30

Do not write below this area!

The following information is used in the next three questions. Consider a market with the following demand and supply schedules:

$$Q^d = 200 - 10p^d$$

$$Q^s = 20 + \frac{10}{3}p^s$$

Since the only things in life that are certain are *death and taxes*, the government needs to levy a tax on this market in order to raise revenues. However, the government has two different policy instruments in order to implement the tax: *per-unit taxes* and *ad-valorem taxes*. Per-unit taxes are similar to the ones we studied a tax of t per unit introduces a wedge $t = p^d - p^s$. On the other hand, the ad-valorem tax is over total expenditures of the good, i.e., this tax does not apply to each unit but to the total expenditure. Thus, an ad-valorem of τ satisfies: $p^d = (1 + \tau)p^s$.

1. Suppose that the government puts a per-unit tax of $t = 2$. What would be the ad-valorem tax τ that generates the exact same level of government revenues as this per-unit tax?

- (a) $\tau = \frac{4}{15}$
- (b) $\tau = \frac{1}{2}$
- (c) $\tau = \frac{5}{3}$
- (d) $\tau = \frac{1}{6}$

2. Suppose that the government puts a per-unit tax of $t = 3$. What would be the ad-valorem tax τ that generates the exact same level of government revenues as this per-unit tax?

- (a) $\tau = \frac{1}{6}$
- (b) $\tau = \frac{1}{2}$
- (c) $\tau = \frac{5}{3}$
- (d) $\tau = \frac{4}{15}$

3. Now consider the dead-weight loss generated by each of the cases above. You can conclude that

- (a) For a per-unit tax of $t = 2$, the per-unit tax creates a smaller dead weight-loss than the corresponding ad-valorem tax; for the per-unit tax of $t = 3$, the per-unit tax creates a larger dead-weight loss than the corresponding ad-valorem tax.
- (b) For a per-unit tax of $t = 2$, the per-unit tax creates a larger dead weight-loss than the corresponding ad-valorem tax; for the per-unit tax of $t = 3$, the per-unit tax creates a smaller dead-weight loss than the corresponding ad-valorem tax.
- (c) For a per-unit tax of $t = 2$, the per-unit tax creates an equal dead weight-loss than the corresponding ad-valorem tax; for the per-unit tax of $t = 3$, the per-unit tax creates an equal dead-weight loss than the corresponding ad-valorem tax.
- (d) For a per-unit tax of $t = 2$, the per-unit tax creates a larger dead weight-loss than the corresponding ad-valorem tax; for the per-unit tax of $t = 3$, the per-unit tax creates a larger dead-weight loss than the corresponding ad-valorem tax.

Consider an industry that has free entry and exit. Suppose that initially it is in a long-run equilibrium with constant costs with a fixed number of firms, and zero profits. Furthermore, suppose that firms are identical, and that they have upward sloping marginal costs. With this information, answer the following questions

4. In the short-run, with a fixed number of firms, the supply curve is:

- (a) Upward sloping
- (b) Vertical and perfectly inelastic
- (c) Horizontal and perfectly elastic
- (d) None of the above.

5. In the long-run, the long-run supply curve is:

- (a) Horizontal and equal to the minimum average cost.
- (b) Downward sloping and equal to the minimum average cost which is decreasing.
- (c) Upward sloping and equal to the minimum average cost which is increasing.
- (d) I don't know because I should have studied more.

Now suppose that the government applies a quantity tax on this industry.

6. After the adjustment to the new equilibrium in the short-run:

- (a) Consumers will face a higher price, and producers will receive a lower price, compared to the original equilibrium price; producers will be making economic losses.
- (b) Consumers will face a higher price, and producers will receive a lower price, compared to the original equilibrium price; producers will be making economic profits.
- (c) Consumers will face a lower price, producers will receive a higher price, compared to the original equilibrium price; producers will be making economic profits.
- (d) Consumers will face a higher price, producers will receive the same price as they were receiving before the tax; producers will be making zero profits.
- (e) Consumers will face a lower price, producers will receive a higher price, compared to the original equilibrium price; producers will be making economic losses.

7. The imposition of the tax will lead some firms to:

- (a) Leave the industry since they will be making losses.
- (b) The same number of firms to remain in the market because profits do not change in the short-run.
- (c) Enter the industry since they will be making profits.

8. In the long-run:

- (a) Upward sloping because the tax has the effect of restricting quantities.
- (b) Is horizontal and equal to the minimum average cost.
- (c) Downward sloping due to the imposition of the tax.
- (d) Upward sloping due to the imposition of the tax.

9. In the long-run, consumers will end up paying:

- (a) The full amount of the tax.
- (b) A greater proportion of the tax compared to what they were paying in the short-run, but not the full amount.
- (c) A lower proportion of the tax compared to what they were paying in the long-run.
- (d) The same amount of the tax as they were paying in the short-run.

The following information is used in the next six questions. Consider a competitive industry that operates in three markets: Astana, Karaganda, and Semey. Each market can be described by the demand equations:

$$q_D^A = 200,000 - 2P,$$

$$q_D^K = 200,000 - 4P,$$

$$q_D^S = 200,000 - 6.5P.$$

There are 4300 identical firms with the following cost structure for both the short-run and the long-run: the total

cost is $C(q) = 11,000 + 2100q + 450q^2$, and the marginal cost is $MC(q) = 2100 + 900q$.

10. What is the shutdown price of each firm?
 - (a) 6549.72
 - (b) 4.94
 - (c) 2100
 - (d) 0

11. What is the market equilibrium price in the short-run?
 - (a) 38,044.3
 - (b) 23,927.8
 - (c) 30,988.5
 - (d) 35,307.4
 - (e) 18,623.6

12. What is the market equilibrium quantity in the short-run?
 - (a) 78,946.3
 - (b) 171,734
 - (c) 188,156
 - (d) 104,289
 - (e) 214,069

13. What is the long-run equilibrium market price?
 - (a) 4.94
 - (b) 6549.72
 - (c) 0
 - (d) 2100

14. What is the long-run equilibrium market quantity?
 - (a) 518,129
 - (b) 186,901
 - (c) 360,702
 - (d) None of the above

15. How many firms operate in the long-run?
 - (a) 104,797
 - (b) 104,798
 - (c) 104,796
 - (d) None of the above

The following information will be used for the next four questions. Everybody knows that Prof. A (where A is for Anonymous) likes his cat. One student, Sabina, gave a recipe for cat food to Prof. A. At first, Sabina was planning to provide a recipe for pancakes for cats. However, pancakes can be harmful to their health. Cats are obligate carnivores, which means that they require a diet rich in animal protein to thrive. Pancakes are high in carbohydrates and low in protein. Therefore, Sabina provided the recipe for a vet-approved, protein rich food. The main ingredients are 74 g of baked chicken breast c , 46 g of cooked quinoa q , and 23 g of boiled spinach s for 3 portions of the meal. The price of 1 kg of chicken is ₹2500, the price of 1 kg of quinoa is ₹5000 and the price of 1

kg of spinach is ₸1200. Let f denote the quantity of this meal.

16. According to the above recipe, what would be the appropriate production function to model the recipe that Sabina provided to Prof. A?

- (a) $f = 3(74c + 46q + 23s)$
- (b) $f = 3 \min\{\frac{c}{74}, \frac{q}{46}, \frac{s}{23}\}$
- (c) $f = 3c^{1/74}q^{1/46}s^{1/23}$
- (d) $f = \min\{\frac{c}{74}, \frac{q}{46}, \frac{s}{23}\}$

17. For the production function modelling the recipe, what kind of returns to scale it exhibits?

- (a) Constant returns to scale.
- (b) None of the above.
- (c) Increasing returns to scale.
- (d) Decreasing returns to scale.

18. What is the cost function $C(f)$ for Sabina's recipe?

- (a) $C(f) = 442.6f$
- (b) $C(f) = \frac{2213}{15}f$
- (c) $C(f) = \frac{f}{3}(185)^{1/74}(230)^{1/46}(138/5)^{1/23}$
- (d) $C(f) = \frac{5}{444}f$

19. What would be the per-unit cost of one portion of the meal?

- (a) 442.6
- (b) 0.47
- (c) 0.01
- (d) 147.53

The following information is used in the next two questions. In the market for nutritional bars there are two types of consumers, men and women. Suppose that the market equilibrium happens at the same price and quantity for both the women's and men's market. At the *market equilibrium*, the price elasticity of demand for men is $\epsilon_P^M = -2$, and the price elasticity of demand for women is $\epsilon_P^W = -1.7$. Suppose that the supply for nutritional bars in this market is perfectly elastic. Furthermore, suppose that the government imposes a subsidy of ₸5 per unit in this market.

20. Fill in the blank: The increase in the consumer surplus for women will be _____ compared to the increase in the consumer surplus for men.

- (a) greater
- (b) lower
- (c) equal

21. The dead weight loss will be greater in which market?

- (a) the same in both markets
- (b) men
- (c) women

The following information will be used for the next four questions. The market for wheat in Kazakhstan is

described by the following *inverse* demand and supply schedules:

$$P_w = 400 - 0.44Q_w$$

$$P_w = 2Q_w$$

where P_w is the price per bushel of wheat and Q_w are bushels of wheat, in tons.

In order to help wheat producers in Kazakhstan, the government promises to buy the whole crop of the year at \$350 per bushel.

22. At the price of \$350, producers offer the amount:

- (a) 175
- (b) 163.934
- (c) 113.636
- (d) None of the above

23. The per-unit subsidy that the government needs to implement to be able to sell all the wheat bushels at the \$350 price is:

- (a) 28
- (b) 26
- (c) 27
- (d) 25

24. The total subsidy (government expenditure) is:

- (a) 4550
- (b) 4725
- (c) 4375
- (d) 4900

25. Consumer surplus after the subsidy is:

- (a) 6913.66
- (b) 6737.5
- (c) 6392.08
- (d) 6563.62

The following information will be used for the next five questions. After the success that was Sabina's cat food recipe, she tells Prof. A that there are cat cafes in Astana, where customers can buy food for their cats, coffee for themselves, or just play and have fun with the cats that live there. After learning about this, Prof. A gets interested and starts thinking about opening a cat cafe in Astana. Since Sabina just passed her BUS101 course with flying colors, Prof. A decides to partner with her in this new business venture. However, Sabina, having taken ECON101 before, understands the very important need to make a detailed economic analysis.

It turns out that the production function of a cat cafe can be modeled as follows: Let h be the total number of happy customers served in the cafe, then we have that

$$h = F(L, K, Z) = L^{2/3} K^{1/3} Z,$$

where L is hours of labor from the cat cafe attendant, K is the number of cats in the cafe, and Z is a variable that can either be 1 or 0, and denotes whether the fixed costs of the cafe are paid (rent/lease, renovation, permits, etc.); notice that production cannot happen unless these fixed costs are paid.

Sabina estimates that rent per month, plus renovation, furnishing, and monthly repair costs, plus licensing and permits would amount to a total of ₸700,000 per month, i.e., $p_Z = 700,000$. In order to determine the other costs, Sabina and Prof. A decide to hire Daniyar as a consultant, who is an authority in *creative accounting* as well as a certified cat expert. Daniyar estimates that the monthly cost of maintaining a single cat (dry food, wet food, veterinary care, grooming, insurance, basic needs, and toys) would be around ₸100,000 per cat, per month, i.e.,

$p_K = 100,000$.¹ Sabina and Prof. A also decide to hire Daniyar as the attendant of the cat cafe, since he loves cats and is very good at handling difficult clients. They decide to pay Daniyar a wage rate per hour of ₹5,000, i.e., $w = 5,000$.

26. Suppose that Prof. A and Sabina decide to go forward, and *sink the investment* of paying the cost of Z , and start the cat cafe with $K = 8$ cats. If they want to serve h customers, what is the short-run total cost function of the cafe?

- (a) $SRTC(h) = \frac{2500}{\sqrt{2}}h^{3/2} + 1,500,000$.
- (b) $SRTC(h) = \frac{2500}{2\sqrt{2}}h^{3/2} + 1,500,000$.
- (c) $SRTC(h) = \frac{5000}{\sqrt{2}}h^{3/2} + 700,000$
- (d) $SRTC(h) = \frac{2500}{\sqrt{2}}h^{3/2} + 700,000$

27. Being an excellent ECON101 student, Sabina recognizes that the decision to open the cafe should also be made before deciding to sink any investment, i.e., they need to solve the long-run problem. If they want to serve h customers, what is the long-run total cost function of the cafe? You will need the following: if $Z = 1$, $MP_L = \frac{2}{3}\left(\frac{K}{L}\right)^{1/3}$, and $MP_K = \frac{1}{3}\left(\frac{L}{K}\right)^{2/3}$.

- (a) $LRTC(h) = 15,000(5)^{1/3}h + 1,500,000$ for $h > 0$, $LRTC(0) = 0$.
- (b) $LRTC(h) = 15,000(5)^{1/3}h + 700,000$.
- (c) $LRTC(h) = 15,000(5)^{1/3}h + 700,000$ for $h > 0$, $LRTC(0) = 0$.
- (d) $LRTC(h) = 1,500,000$ for all $h \geq 0$.

¹You may think this is very expensive, but Prof. A instructed Daniyar to estimate this number based on cats that are treated as royalty, as it should be.